THP Jump Training Ebook



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Welcome to the THP jump training ebook. I wrote this to help you jump higher. It compiles everything I know about the science behind what I consider the five pillars of jumping training:

1. Strength training
2. Plyometrics
3. Periodization
4. Managing tendon pain
5. Jump technique

If you can implement those five concepts into your training, you WILL jump higher. But not just that. You'll be able to continue making progress year after year while staying healthy. This is the formula to crack through any vertical jump plateau.

At THP we are concerned with one thing. Reaching your genetic potential when it comes to the vertical jump. Depending on genetics, everyone has a different ceiling in terms of jumping ability. However, there is only one way to find out what that ceiling is. Training extremely hard, extremely smart, for years.

Training smart means knowing what to do, in what way to do it, and when to do it.

And the principles that I go over in the next few pages are at the foundation of training smart. The version of you that understands these principles will improve more than the version that doesn't. Even if those two versions of you were to do the exact same training plan.

Why is that?

Because understanding the science helps you train correctly.

For example, if you know about the principle of progressive overload, you know that training must get harder week to week. So you will keep increasing intensity week to week. And if you know about periodization you will understand that it's normal for your vertical to go down when training hard. And you will also know to back off when you're supposed to in order to achieve supercompensation.

I hope that you can get value out of this guide. Enjoy!

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The Fundamentals Of Strength Training

Goal: Understand what strength is, why it is important when it comes to jumping higher, and how we choose exercises in a training program.

Five biomotor capacities

1. **Strength**
2. Speed
3. Flexibility
4. Endurance
5. Coordination

What is adaptation?

- Adjustment of an organism in response to its environment
- If environment changes, organism changes to survive
- Exercise/physical activity is a powerful stimulus for adaptation
- The goal of exercise is to induce adaptation

The four features of adaptation

1. Overload aka stimulus magnitude
2. Accommodation
3. Specificity
4. Individualization

Overload

- Adaptation only occurs if training load is above habitual level
- Two ways to overload
    - Increase training load through intensity or volume
    - Variety: Changing the exercise
- Three types of training loads
    - Stimulating: Adaptation
    - Retraining: Maintenance
    - Detraining: Performance decrease
- The more elite the athlete, the easier it is to detrain, thus a higher training load is needed

Accommodation

- The response of a biological object to a constant stimulus decreases over time
- The principle of diminishing returns

Specificity

- Training adaptations are highly specific
- The more similar a training stimulus is to the activity you are trying to improve, the more transfer the stimulus will have

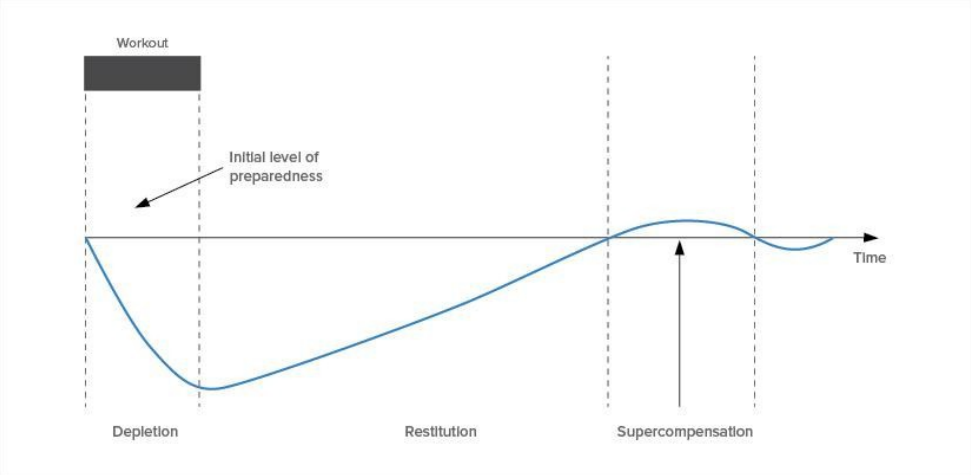
Individualization

- All people are different

Supercompensation & fitness/fatigue model

Terms to understand:

- Preparedness
  - How well equipped an athlete is to handle a training session or competition
  - Affected by training fatigue, psychological overstress, illness
- Fitness
  - An organism's ability to survive its environment, which in our case is training!
  - Think of it as long term preparedness
- Fatigue
  - Decrease in physical performance due to exercise
  - Affects your preparedness
- Fitness-fatigue model determines supercompensation curve
- Each biomotor capacity has its own supercompensation curve



Effects of training

- Changes that occur within the body as a result of training
  - Acute effects: Happen during exercise

    - Immediate effects: Manifested soon after working out
    - Cumulative effects: Occur as a result of continued training sessions
    - Delayed effects: Manifest over a given time interval after exercise
    - Partial effects: Changes produced by a single exercise
    - Residual effects: Retention of changes after cessation of training during which **adaptation** can take place

What is strength?

Strength

- The ability to generate maximum external force

Force

- Instantaneous measure of the interaction between two objects
- Characterized by
    - Magnitude
    - Direction
    - Point of application
- Human body
    - **External: Athlete's body vs. environment**
    - Internal: bone on bone, tendon to bone, etc.
    - Muscle exerts forces on bone while
        - Shortening
        - Lengthening
        - Remaining the same length

Why is strength training important for increasing your vertical jump?

- Strength training improves your ability to generate force
- Your bodyweight provides high resistance

PS. If you want my team to program a training cycle specific to your needs based on these principles sign up for our FREE 6 week challenge HERE

The role of resistance and extrinsic factors

- Different types of resistance
    - Inertia
        - Law of inertia: An object at rest stays at rest unless acted upon by an outside force

        - F=ma
        - In jumping, your body's mass is the parameter. So the force you apply on it determines how fast you can accelerate it upwards, which determines how high you jump
    - Elasticity
        - F=kD
        - Force, coefficient of stiffness, displacement through deformation
        - Bands, springs, stretch cords
    - Hydrodynamic resistance
        - Swimming, rowing, kayaking
    - Compound resistance
        - This is when there a multiple forms of resistance
        - Bands + weights
            - Overcome barbell weight, barbell inertia which is proportional to acceleration of barbell, and elastic force

Force Constraints

- Time available for force development
    - Time to peak force varies with each person and with different motions
    - Typically when measured isometrically, time to peak force is 0.3 to 0.4 s
    - Here is how long some track and field movements take:

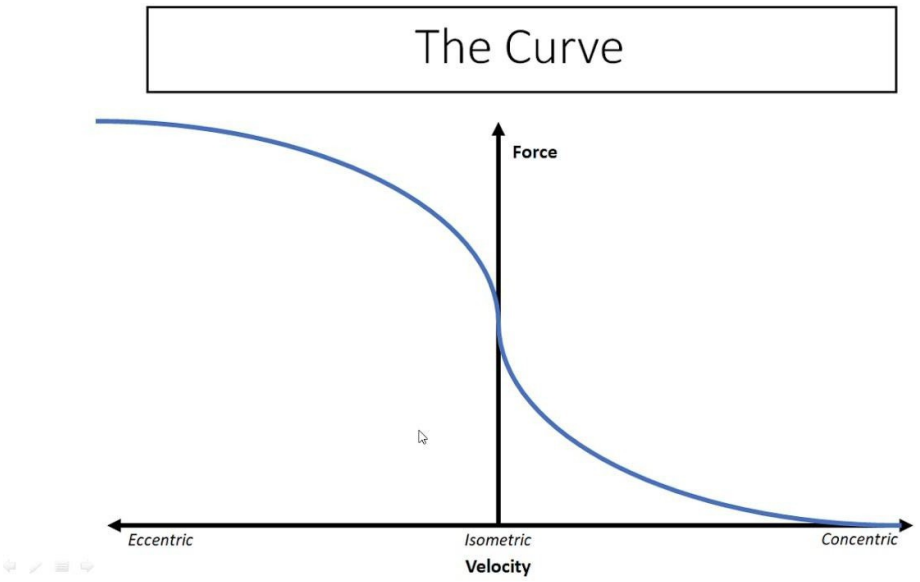
Motion	Time (s)
Takeoff	
Sprint running	0.08-0.10
Long jump	0.11-0.12
High jump	0.17-0.18
Delivery	
Javelin	0.16-0.18
Shot put	0.15-0.18
Hand takeoff (horse vaulting)	0.18-0.21

- In jumping and sprinting, the time available to complete the movement is less than than the time needed to get to the maximal possible force in ideal conditions
- As resistance decreases, you're able to move faster, so the difference between peak force and peak possible force in the most ideal conditions increases. This difference is called the **explosive-strength deficit**
    - The explosive strength deficit shows the percentage of an athlete's strength potential that was not used in a given attempt

- **Explosive strength** is the ability to exert maximal forces over a minimal time
- There are two ways to increase explosive strength
  - Increase max force capabilities (typically most effective for novices)
  - Decrease explosive strength deficit by improving rate of force development
  - When an athlete improves jump technique, they're able to put themselves in a position where they can apply more force and they're also able to complete the jump in less time. This means that RFD plays a bigger role as the athlete becomes more advanced

The force velocity curve

- Is the interplay between force and velocity with time as a constraint in muscle actions
- In concentric muscle actions, motion velocity decreases as external resistance increases



- Consequences of the force velocity equation in sports
  - It is impossible to exert high force in very fast movements
    - In a power clean, the position where a lifter is strongest in a pull is around knee height. If there is too much initial velocity there will be less force able to be generated when the bar reaches knee height, so you essentially miss your golden opportunity to accelerate the bar upwards.
  - Forces applied in the intermediate ranges of the curve are very dependent on an athletes max strength

        - This means that an athlete's maximal strength determines the force that can be exerted dynamically, which is what we care about in jumping (more reasons why getting stronger is so important for jumping!)
        - Now the slower the speed of the movement, the more related performance will be to max strength
        - In extreme areas of the curve, this correlation falls off
    - Maximal power is achieved in the intermediate range of force and velocity
        - Generally this is around ⅓ of max velocity and ½ of max force, with it varying based on the movement

- Eccentric muscle contractions
    - Forces in the yielding phases of motion during stretch shortening cycles can easily exceed the maximal isometric strength of an athlete
    - Eccentric forces substantially increases with initial increase in movement velocity and the velocity of muscle lengthening
- We will go over stretch shortening cycles more in depth in the plyometrics section of the jump training course

The role of torque in jumping

- How much force you can apply is also related to your body posture in a lift/jump and limb lengths
- The farther away the point of force application is from the axis of rotation, the more torque is required to apply force to a resistance ie. deep squatting vs quarter squatting
- Longer limbed athletes also have to generate more torque in order to apply the same force as a shorter limbed athlete
- The more torque you can apply, the higher you'll be able to jump!

Peripheral and neural factors in force application

Peripheral factors

- Capabilities of individual muscles
- Capabilities of the muscle are largely influenced by your muscle dimensions
    - Muscles with larger cross sectional area generally produce higher forces
    - Cross sectional area increases with proper training and nutrition
    - Muscle size growth occurs from hypertrophy and NOT hyperplasia
    - The way we grow muscles is through resistance training. Heavy resistance training breaks down muscle proteins, creating conditions for synthesis of contractile proteins during rest periods

    - Generally athletes with greater body weight have greater levels of absolute strength, and athletes with lower body weights have greater levels of relative strength
        - Because of this, managing weight loss correctly can sometimes increase your relative strength
        - Relative strength is what is important for jumping higher!!
- Neural aka central factors
    - This is coordination of the muscles by the CNS
    - Intramuscular coordination
        - How individual muscle fibers are innervated
    - Motor units
        - Motor units are motor neurons connected to muscle fibers
        - Slow motor units are used in activities with slow velocity and vice versa for fast motor units
    - How do motor units coordinate muscles?
        - Recruitment of motor units (how many)
        - Rate coding (how quickly)
        - Synchronization of motor units
        - Motor units are recruited through the size principle(small first)
    - Intermuscular coordination
        - How different muscle groups coordinate with each other
        - The complexity of intermuscular coordination is why compound movements have better transfer to jumping higher… jumping is complex!!

- Elite athletes have better intra and intermuscular coordination

Types of muscular strength

Static strength
- Isometric and slow concentric actions

Dynamic strength
- Fast concentric actions

Yielding strength
- Eccentric actions

Explosive strength and SSC's are independent components of motor function that rely more heavily on tendons

*These are not all inclusive, and in real life there are hybrids and smooth transitions between all of these types of strengths

Methods of strength training

How exactly do we put all this information together to make a training program that is going to make you jump higher?

- We choose the exercises and put them together in a training plan in a way that causes long term adaptation, and continues to spur on adaptation year after year!

Variables that we manipulate:

- Training Intensity
    - Single vs multi joining
    - Velocity of movement
    - Total load
    - How close to failure
    - Intent

- Volume
    - How much work you are doing
    - Sets/reps*load
    - Distance

How we manipulate these variables in order to have an athlete peak in performance at certain times of the year and continue making long term progress is called **periodization**

- Sets/reps
- Deload weeks
- Exercise selection
- Defining different types of training cycles

How we periodize the training based on these variables will be covered in the periodization section of the jump training course

Types of exercises

We break exercises down by:

1. Muscle groups
    - Posterior chain

- Knee extensors
- Lower leg
- Core
- Upper body
    - Pulling
    - Pushing
    - Shoulders

2. Where they fall on the force velocity curve
    a. We want to surf the force velocity curve from right to left in eccentric muscle actions

Injury Prevention

Load management considerations

- Type of injury
- Severity of injury
- Sports
- Goals

Plyometrics and the stretch shortening cycle

Forces in the yielding phases of motion under conditions of imposed muscle lengthening are called eccentric or plyometric muscle actions

- These can easily exceed the maximal isometric strength of an athlete

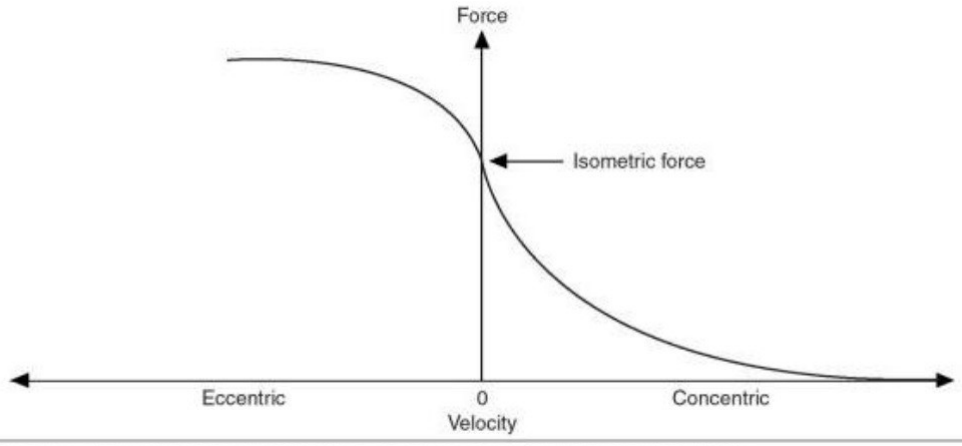


Figure 2.14 Force–velocity curve for concentric and eccentric muscular actions.

The stretch shortening cycle aka reversible muscle actions

Stretch shortening cycle

- Movements that consist of an eccentric(stretching) phase and a concentric(shortening) phase

When a muscle shortens immediately after a stretch:

- Force and power output increases
- Energy expenditure decreases

How the SSC augments force production and decrease energy expenditure

1. At the transition from eccentric to concentric, force is developed in an isometric condition. Because of this, the influence of high velocity is avoided and you can tap into your maximum possible force (refer to the force velocity curve when velocity is at zero)
2. Since force begins to rise in an isometric phase, the time available for force development is greater (ie. paused squat jumps has no isometric phase vs. countermovement jumps gives more time to apply force through the isometric phase)
3. Muscle and tendon elasticity
    a. When a tendon and muscle is stretched, elastic energy is stored
    b. This deformation energy is recoiled and used to enhance motor output in the concentric phase of the SSC
    c. Magnitude of stored energy is proportional to applied force and induced deformation
    d. Muscles and tendons are arranged in series, so they are subjected to the same force and the distribution of stored energy between them is only a function of their deformation
    e. The deformation is a function of muscle and tendon stiffness or it's inverse value, **compliance**

f. The stiffness of a tendon is constant, while the stiffness of a muscle is variable and depends on the forces exerted
g. A passive muscle is compliant aka it can be easily stretched and an active muscle is stiff
h. It takes great force to stretch a stiff muscle
i. Superior athletes can develop high forces and the stiffness of their muscles while active can exceed the stiffness of their tendons. What is the implication of this? A sh*t ton of elastic energy is now stored in the tendon because they are extremely hard to stretch!
j. This is why elastic energy in elite athletes is stored primarily in the tendons rather than in the muscles

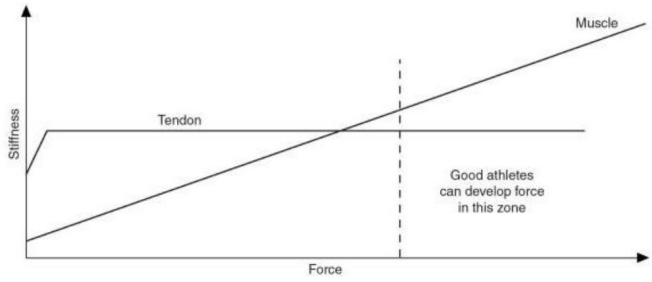


Figure 2.17 Stiffness of a muscle and a tendon at different levels of muscular force. Since elite athletes develop high forces, the stiffness of the muscle, while active, may exceed tendon stiffness. In such cases, the tendons are deformed to a greater extent than the muscles and thus store more potential energy.

4. Neural mechanisms
    a. During stretching phases of the SSC, there is a rapid change in both muscle length and the forces developed. Muscles are forcibly stretched and at the same time muscle tension rises sharply. These changes are controlled and partially counterbalanced by two motor reflexes
    b. Myostatic or stretch reflex
        i. keeps muscles close to preset length
        ii. Muscle spindles are arranged parallel to the muscle fibers. When muscle is stretched by external force the muscle spindles are also stretched. This stretching induces a discharge of alpha-motorneurons that causes a reflex contraction of the stretched muscle
    c. Golgi tendon reflex
        i. prevents unusually high and potentially damaging muscular tension
        ii. Golgi tendon organs are arranged in series with muscle fibers. The receptors are sensitive to **forces** developed in the muscle rather than length changes. If muscle tension increases sharply, the golgi tendon reflex inhibits muscle action. The ensuing drop in muscle tension prevents the muscle and tendon from incurring damage
        iii. If an athlete(even experienced ones) is not accustomed to a plyometric exercise, the activity of the extensor muscles are inhibited by the golgi tendon reflex

1. Specific training can result in inhibition of the golgi tendon reflex and the athlete would be able to sustain very high landing forces without a decrease in exerted muscular force
2. This is also why HELLA jump reps are so crucial!!!
3. This is also why experienced jumpers would mop the floor with even the most elite powerlifters in depth jump exercises
4. In beginners, just weight training can improve the SSC. But as an athlete becomes more experienced, very specific plyometric exercises are needed to see progress

How do plyometrics increase your vertical?

- An environment conducive to creating adaptation is created by overloading eccentric muscle actions through exercises such as drop jumps, repeat jumps, and overspeed approaches. The adaptation is being able to generate more force in all parts of the SSC
- Plyometrics also self intensify, so as you get better at doing plyos you continually overload your body and keep spurring on adaptation
    - This is why someone can do no weight training and only jump for years and make progress, assuming they can stay healthy. You essentially bypass the law of accommodation, until you start reaching elite levels of jumping
- Makes your muscles and tendons able to store more elastic energy
- Improves the stretch reflex of muscles
- It inhibits the golgi tendon organ response

By the way! We have built over 250 unique weeks of training each with their own plyometric progression. To apply the one that best fits your specific needs sign up for our FREE 6 week challenge HERE

Planning Plyometrics For Your Training

Progressive Overload

- Progressive overload is used to induce adaptations
- In plyometrics the goal is to improve power development
    - Progressive overload is achieved through exploiting the equation for power
    - Side note: Jump height is directly 1 to 1 related to impulse (a change in momentum). However there is a correlation to power, and it is easy to plan plyometrics when looking at the equation to power
- Power = Fxd/t
- Where F=force, d=distance, and t=time

- With this in mind, here is how we can overload plyometrics within each variable of the power equation

1. Resistive overload
    a. Gravitational
        i. Falling from a height
    b. Inclination
    c. External
2. Spatial overload
    a. Range of motion
        i. Move limbs through much wider or much shorter ROM's
        ii. Start from a standstill instead of an approach
    b. Sagittal, transverse, and frontal planes of motion
        i. Split lunge
3. Temporal overload
    a. Operating rate
        i. Executing movements with more intent and more speed
    b. Impulse

Specificity

- Exercise creates specific neuromuscular and metabolic adaptations to overload
- Transfer to your sport depends on how similar the plyometric is to the activity you're trying to improve

Evaluating exercise objectives

- SSC appears everywhere
    - Warm up, weight room, speed portion, agility portion
- Sometimes SSC will be used specifically, and other times generally

Time lapse for the SSC

- Minimizing the time lapse between the eccentric and concentric contraction is extremely important
- Two periods of delay exist
    - Between signal from brain for muscle contraction and onset of muscle activity
    - Between appearance of muscle electrical activity and development of tension in the muscle
        - This delay is shorter in eccentric contractions which supports concept of producing the greatest force in the least amount of time

        - SSC enhances force production from the series elastic component (SEC) during stretching AKA eccentric efficiency is improved using the stored elastic energy of the SEC

SSC factors that determine timing, volumes, and intensities of plyometrics

- Rate of eccentric action aka amortization phase aka the stretch
- Rate of concentric action aka the recoil aka the summation phase aka shortening
- Delay time between cessation of eccentric and onset of concentric muscle action aka coupling time
- Amount of external load involved

Different types of SSC movements

- Intensive elastic and reactive method
- Speed strength (loaded and not as elastic or reactive)
- Shock methodology
    - If coupling time is longer than 0.15 seconds during high impact/intensity/RFD drills than it's NOT classical shock-method plyometrics
- These can more simplistically be defined as short vs long SSC's
    - Short: < 250 ms
    - Long > 250 ms

Which plyometrics should you do?

- Plyometrics fall along a large continuum of exercise and load factors
- Evaluate specific goals for each training phase, and this goal will tell you where on the continuum you should choose plyometrics from

Volume and dosage

- Number of repetitions are usually 8-12
- Eastern Europe scientists suggest 6-10 sets
- Russian sport scientists suggest 3 to 6 sets

Single and multi response drills

- Two categories for SSC drills
    - Single response
        - Single intense effort
    - Multi response
        - Several efforts in succession
        - More emphasis on elasticity, speed, and coordination
        - One huge goal of plyometric training is accomplishing high-impulse landings and takeoffs in succession
    - There's also a third type
        - Multi response with pauses
- Both responses should be used to properly progress through the plyometric training continuum

- Objective of drills
    - Improve nerve-muscle reactions, explosiveness, quickness, and ability to generate forces in certain directions
    - An athlete will only benefit from the number of repetitions that are done well, this includes intent and proper execution!

Intensity

- Described in two ways in regards to SSC
    - Force at impact
    - Intent or level of effort
        - Quickness of execution is essential for optimal training effects
        - Rate of muscle stretch is more important than the magnitude of stretch
        - Greater reflex response occurs when you load the muscle rapidly which is done by using maximum effort

Maximize force and minimize time

- Both force and velocity are important
- In many cases the critical concern is the speed at which you can perform a particular action
    - Example: shot put
        - The quicker you execute the action sequence, the greater the force you will generate and the longer distance you will achieve
- Impulse of the movement is key
    - Movements must have high impulse for plyometric training to be effective

Guidelines for workouts and microcycles

- Rest periods
    - 1-2 minutes is usually enough rest for the neuromuscular systems stressed by SSC exercises to recuperate
    - Low impact and low intensity plyos may require minimal rest of 30-60 seconds
    - At the shock end of the plyo stress scale exercises may require 2-3 minutes of rest for systems to be able to handle and produce forces necessary for optimal execution
- Frequency of loading
    - 2-3 days per week usually gives optimal results
        - Consider total training load, type of activity, and inverse relationship between frequency and intensity
- Order of exercises
    - Dynamic and more intense first

Complex training

- Pairing strength exercises with plyos that have similar movement patterns

Types of jumps

- Jumping
    - Landing on both legs
- Squat jumps
    - From static position
- Counter movement jump
    - Using a pre stretch by flexing at ankle, knee and hip and then extending
- Depth (drop) jump
    - Jump after dropping from height
- Bounds
    - Takeoff leg to leg
    - Can also be double leg to double leg
    - Usually Emphasis on max horizontal distance
- Hopping
    - Same leg to same leg
    - Either for distance or height
- Leaps
    - Single effort exercise emphasizing max horizontal or vertical distance
- Skips
    - Alternating step hop
    - Left-left to right-right to left-left to etc.
    - Emphasis on max height or distance
- Ricochet
    - Emphasis is purely on fast leg and foot movement

Choosing depth jump height

1. Test your standing vertical
2. Perform depth jumps starting with a low box
3. Increase height progressively until you test a lower vertical than your standing jump
4. Perform depth jumps at the the height where the depth jump matches your standing jump height

PS. Plyos are one of the hardest parts of training to get right. They're perfect for some athletes… or they could make other athletes jump lower! That's why coaching is so important. At THP you'll always know you are doing the right training for YOU. Sign up for our FREE 6 week challenge HERE

PERIODIZATION

The deliberate manipulation of training variables to enhance sport performance at specific times. The main obstacle is timing the training correctly while dealing with an athlete's schedule

Structural units of training and definitions

- Training session
- Training day
    - Can consist of multiple training sessions that are spaced by more than 30 minutes of rest
- Microcycle
    - Grouping of several training days
    - Usually one week
- Mesocycle
    - System of several microcycles
    - Usually 4 weeks
    - Range of 2-6 weeks
    - Types of mesocycles
        - Accumulative: Enhance athlete's potential
        - Transmutative: Increase non specific fitness into specific athlete preparedness
        - Realizational: Planned to put on best sport performance attainable within given range of fitness
- Macrocycle
    - One entire competition season
    - Includes preseason, in season, and post season periods
        - Aka preparation, competition, and transition periods
    - Usually lasts one year for winter sports, and half a year for track(and dunking!)
    - Olympic macrocycle is four years in length
- **Short term planning**
    - Planning workouts, training days, microcycles, and mesocycles
- **Medium term planning**
    - Planning macrocycles
- Long term planning
    - Deals with training intervals of many years

Short term planning

Principle of fatigue specificity

- The main influencing factor in short term planning is fatigue
    - Fatigue effects from different types of muscular work are specific. **One's ability to execute drills of another type is restored more quickly than one's ability to repeat the same routine**
    - If athlete is too tired to repeat the same exercise in an acceptable manner, they may still be able to perform another exercise to satisfaction
    - Sequencing drills appropriately makes it **possible to assign more work and increase training volume**
    - The same applies for exercises of different directions like sprints vs. strength work
    - **If two similar training workouts are executed in a row, the traces of fatigue from the two sessions are superposed (see image)**

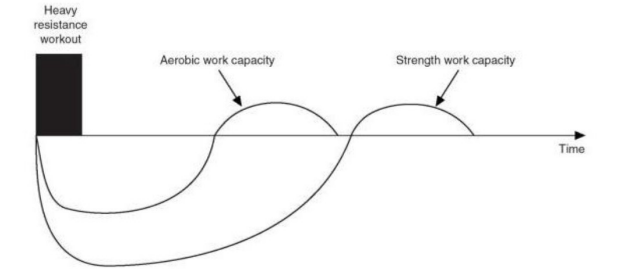


Figure 5.1 The time course of athlete restoration after a heavy resistance training session.

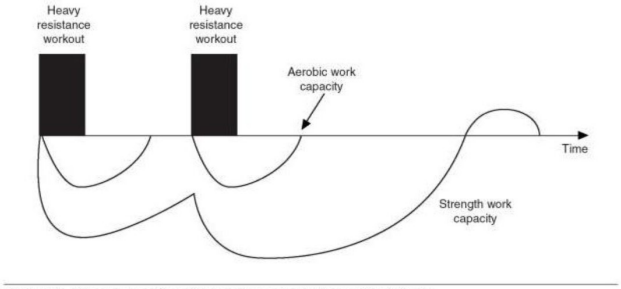


Figure 5.2 The superposition of two resistance workouts and their effects.

Optimal number of motor abilities to focus on?

- Fitness gain decreases if several motor abilities are trained simultaneously during one workout, microcycle, or mesocycle
    - It is not a good idea to have more than 2-3 main targets in a microcycle or mesocycle
    - Ie. no reason to train in one microcycle train max strength, explosive strength, aerobic capacity, anaerobic lactacid and alactacid capacities, max speed, and sport technique
        - Organism cannot adapt to so many different requirements at the same time
        - Gains in all these motor abilities would be insignificant compared with gain from development of only one physical quality
- When training targets are distributed over several mesocycles in **sequence**, fitness gains increase
- **To enhance performance, decrease number of targets micro- and mesocycles**
- **BUT to avoid superimposition of fatigue traces from individual workouts and the hazard of staleness, increase the number of targets in micro- and mesocycles**

What is optimal?

- Number of motor abilities or targets to focus on in one mesocycle
    - 1-2
    - And one essential feature of sport technique (in our case jumping higher or your sport)
    - 70-80% of the total work in a mesocycle should be addressed to the development of targeted motor abilities (35-40% per target)

What is the goal in planning strength training sessions?

- Do as much work as possible while having the athlete be as fresh as possible
- Three facets to timing the workouts
    - Rest-exercise alteration
        - Elite weightlifters do 32-45 sets per day (Most intense THP workouts are around 20)
        - Rest 4-5 minutes when working with large weights
    - Exercise sequence
        - Objective is to perform most valuable exercises requiring fine motor coordination and maximal neural output in a rested state
        - Main sport exercises before assistance exercises
        - Dynamic-power exercises before slower exercises
        - Large muscle groups before smaller ones
    - Intensity variation

        - Highest intensity exercises should be performed at the beginning of the workout

Microcycles and Mesocycles

Two main goals

- Allow adequate recovery between exercise periods
- Find proper balance between steadiness of a training stimulus to call forth an adaptation and its variability to avoid premature accommodation and staleness

Adequate recovery

- In microcycle
    - Rest-exercise alteration and proper exercise sequencing alleviate fatigue
    - Greatest training adaptation occurs when muscles are recovered from previous training periods and best prepared to tolerate the greatest overload
    - Elite athletes train 5-6 times per week(like in THP), in these cases proper rotation of exercises in sequential sessions is essential(back to back sessions should not involve the same muscle groups and repeat the same patterns of muscle coordination)
    - Recovery time for heavy resistance training varies with muscle size, the smaller the muscles the more frequently you can train them
    - To increase muscular strength you want to do heavy resistance training at least three times per week
    - To retain strength gains, at least 2 training sessions per week should be scheduled (that last about 30 minutes)

Variability

- Variability in training programs within a micro- and mesocycle is realized through changes in training load and NOT exercise complexes
- One stable complex of exercises should be performed in a mesocycle to elicit an adaptation
    - Complexes usually consist of ten exercises that are distributed among the training days and workouts of one microcycle
    - Each exercise is usually performed at least twice per week
- To avoid premature accommodation
    - Training loads should vary day to day and microcycle to microcycle
    - Rule of 60%

- Training volume of a day with minimal loading in a microcycle should be 60% of the loading of a maximal day in a microcycle

Medium Term Planning (Periodization)

The issues of periodization

- One of the most complex problems in athlete training is finding the proper balance between opposing demands which is difficult to achieve due to so many factors involved

Four issues

1. Delayed transformation of training loads into fitness
2. Delayed transmutation of nonspecific fitness to specific fitness
3. Training residuals
4. The superposition of training effects

Delayed transformation

- It takes time to adapt to training stimulus
- Hard training induces fatigue that accumulates over time
- Period of relatively easy exercise is needed to realize the effect of the previous hard training session aka **period of delayed transformation**
- Adaptation occurs when retaining or detraining load is used after a stimulating load
- Time of delayed transformation lengthens as the total training load and accumulated training fatigue increases

Delayed transmutation

- “Transmuting” the effects of accumulation mesocycles (where you improve your ability in assistance exercises and thus increase your non specific fitness) into specific fitness
- The number of transmutation mesocycles depends on the number and duration of accumulation cycles
- Transmutation and realization mesocycles when considered together as one unit is called tapering
- To properly program these mesocycles at the right times, training residuals must be taken into account

Training Residuals

- The reduction or cessation of training brings about substantial losses in adaptation effects BUT **athletes can sustain acquired training benefits for a specific period of time without extensively training them continually**. The longer you spent training to induce the adaptation, the longer it'll take to detrain that quality

Four factors that determine time to detraining

1. Duration of the immediately preceding period of training (period of accumulation)
2. Training experience of athlete
3. Targeted motor abilities
4. Amount of specific training loads during detraining (or retaining) mesocycles

Superposition of training effects

- Training effects are specialized
    - Methods that induce beneficial adaptation in one motor ability or physiological system may produce negative effects on another ability or system
    - Transfer of either positive or negative effects between two types of training is not necessarily symmetrical ie. max strength helps power, but power doesn't help max strength

Periodization as a trade off

- A good periodization plan is a subtle tradeoff between conflicting demands. The greatest gains in any one direction occur when an athlete focuses on a specific type of training for 1 or 2 mesocycles
- Attribute analogy!!!

Strength training in macrocycles

Variables that are most important in macrocycles

(same ones as mesocycles with the addition of)

1. **Demand for variability in training stimuli**
2. Delayed transformation of a training load
3. Delayed transmutation of nonspecific fitness into specific fitness
4. Training residuals

Variability of training stimuli (mesocycle to mesocycle)

- Variability demands are met by changing training loads and exercises
- Training load variability by itself has limits due to staleness and time constraints, so changing exercises is preferable in long term planning
- When exercises are changed, there is initial period of decreased performance followed by improvement after period of adaptation to new load

Delayed Transformation (within a mesocycle)

- Because of the time delay between an increase in training load and improvement in performance, a period of delayed transformation is needed before important competitions(or in our case, when you want to jump high)
- The length of time of delayed transformation is positively correlated with the amount of training load
- This period of time has decreased sessions per week, number of exercises, and number of sets

Delayed Transmutation (mesocycle to mesocycle)

- As time to important competition decreases, exercises become more and more specific
- This is the delayed transmutation of nonspecific fitness into specific fitness

Training residuals and retaining loads

- Levels of strength can be maintained in season with retaining loads
- Retaining loads
    - Two 30-40 min heavy resistance workouts per week is usually enough
    - The more elite the athlete, the higher the retaining loads

Strength training methods in macrocycles

- Submaximal effort
    - Lifting non maximal loads to an intermediate number of times (NOT to failure)
- Repeated effort
    - Maximal number of repetitions in a set
- Maximal effort
    - Max strength and absolute strength
    - Maximal loading with minimal number of repetitions
- In a macrocycle, training generally progress from submaximal effort focused mesocycles to maximal effort focused mesocycles
    - This is to initially prepare the musculoskeletal system and then improve neural coordination

Types of periodization

Four main groups of periodization methods

1. Sequential
2. Concurrent
3. Conjugate Sequence
4. Combination of all of these

Important definitions to understand for the types of periodization

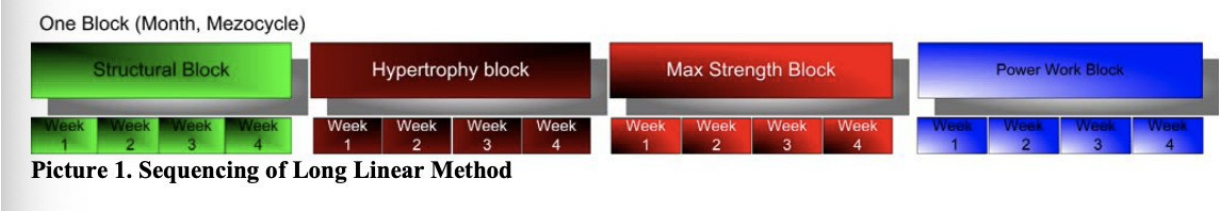
- Long
  - Changes in training goals microcycle to microcycle
- Short
  - Changes in training goals session to session within a microcycle
- Linear
  - Starts with high volume/low intensity and ends with low volume and high intensity
- Undulating
  - Same sequence as linear, but uses waves of inverse volume/intensity within a cycle(workout to workout in short or week to week in long)

Sequential

- Focus on **just one training goal** at a time in sequence cycle to cycle

Types of sequential periodization

- Long linear



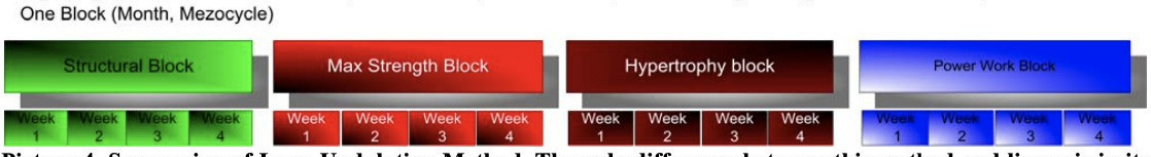
Picture 1. Sequencing of Long Linear Method

- Short linear



Picture 3. Sequencing of Short Linear Method. Note that there is a same sequence as in Long Linear Method but only done in less time.

- Long undulating



Picture 4. Sequencing of Long Undulating Method. The only difference between this method and linear is in its „breaking“ of linear approach, but rather using „waveing“ approach.

- Short undulating



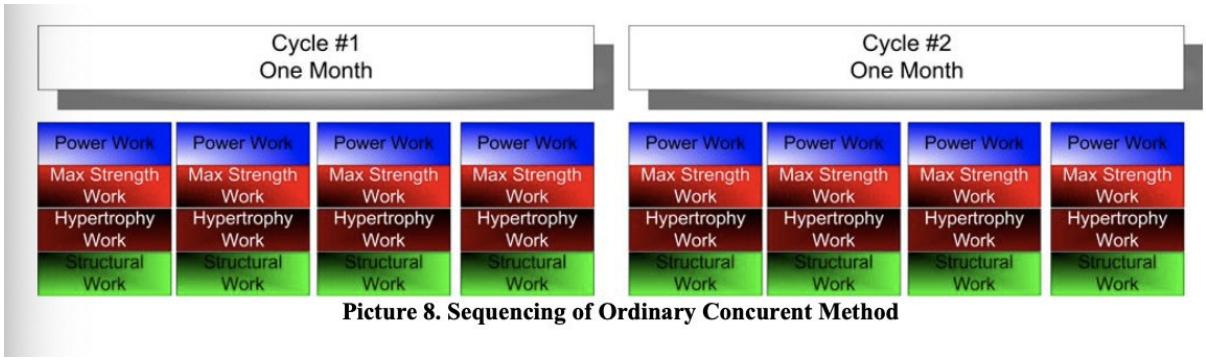
Picture 5. Sequencing of Short Undulating Method.

Concurrent periodization

- **Focuses on all training goals at the same time, with the same amount of emphasis on each goal**

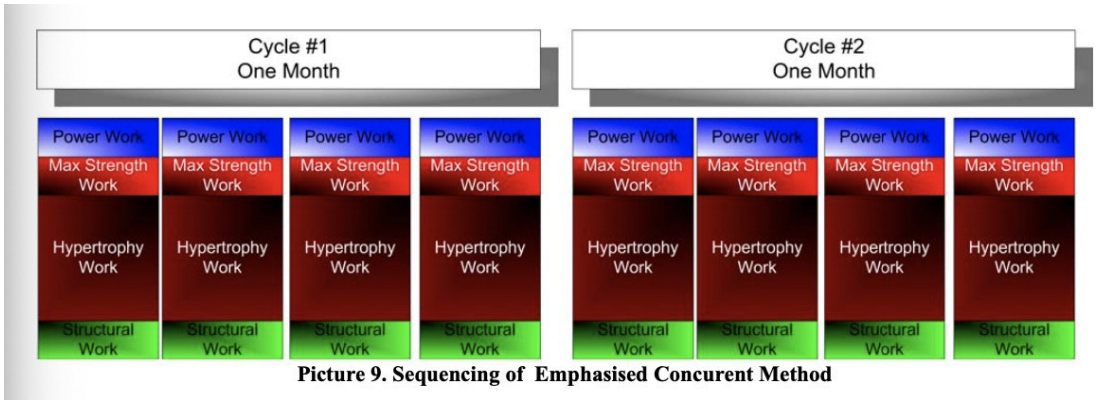
Types of concurrent periodization

- Ordinary concurrent method



Picture 8. Sequencing of Ordinary Concurent Method

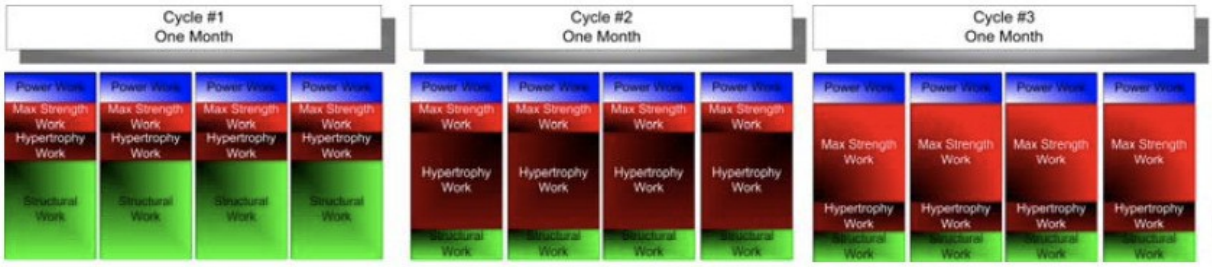
- Emphasized concurrent method



Picture 9. Sequencing of Emphasised Concurent Method

Conjugate Sequence System

- Focuses on all training goals at the same time, with **increased emphasis on 1 training goal and decreased emphasis on all other training goals**



Picture 12. Long Conjugate Sequence System

Periodization is at the heart of every good program. At THP we build periodization into every cycle. You will literally never do the same week of training twice. This guarantees that you are always adapting and pushing to get better. Sign up for our FREE 6 week challenge HERE

TWO FOOT AND ONE FOOT

JUMP TECHNIQUE GUIDE



One of the most important elements to consider in regards to jumping higher is your **jump technique.**

This article will serve to address the main components of an approach vertical jump off of one and two feet. We will discuss the key parts of the approach, their purpose, and how you can optimize them. It is important to understand these terms as it will help you maximize your gains on the court, field, or track.

General concepts

Regardless of if you jump off of one foot or two feet, there are some basic concepts that generally apply to both styles. **The two most important variables that dictate the potential to jump high are how low you are and how fast you are.**

How low you are is defined as the position of your center of mass at touchdown of your plant foot. In other words, how much are you bending at the knee and hip leading into the jump.



Again, this happens in **both** styles of approach jumps. The reason this is so important is that it gives you the potential to generate a large change in **momentum**. As an illustration, imagine trying to throw a baseball with a short wind up, like a catcher, versus a center fielder launching a ball to home plate. Obviously the center fielder would require a much larger wind up in order to throw the ball over 100 yards. This relates back to **impulse** which is defined as the

force you generate over a given time interval. Ultimately, to accelerate your center of mass upwards in jumping, you need to generate a lot of force over a given time interval. The more you lower, the greater the time interval to generate force. However, if you lower **too much** you could risk your leg buckling or being in a detrimental position to generate high intramuscular forces. This is because muscles have an ideal length at which they produce force, as well as your body being in a poorly leveraged position. In other words, your legs could give out.

It is also important to take into account **when you lower**, as this will determine the negative vertical velocity of your center of mass at touchdown of the plant foot. A high negative vertical velocity is detrimental because it will cause you to have a lower jump. For example, in a one foot jump, if I try to lower myself too late my body will be plummeting towards earth right at the instant I am trying to propel myself upwards. In contrast, if I lower early in the run up and have a very fast rhythm, I can minimize this negative consequence. To sum things up, you want to be low to the ground 2-3 steps prior to takeoff.



How fast you are is defined as the velocity of your center of mass at touchdown of the plant foot. This is achieved through the run up and penultimate step, which we will discuss in

greater detail later in this article. Through coaching athletes over the last 15 years, I've added a second definition, which is characterized by how fast the stride frequency is during the approach and takeoff sequence. If you have ever observed athletes who tend to have large braking steps during one foot jumps, it is because they are low and have a lot of momentum (speed), but lack a fast rhythm (also speed). It is important to not only be moving fast, but also to have a very fast rhythm leading into the jump, especially with one foot jumps.

The reason speed is so important is because it allows the athlete to take advantage of the stretch shortening cycle of the musculotendinous unit. Speed can also be used to maximize the deflection of the athletes leg off the ground, but only if they are strong enough to convert the horizontal momentum vertically. If they are not, then their leg will buckle.

All of that being said, being low and fast gives you the **potential** to jump high, but it doesn't mean that you **will** jump high, as the underlying physiology must be able to handle the increased loading on the muscles.

Two foot jumping

In this section, we will discuss in detail the specific portions of **good** two foot jumping so that you can better understand how to optimize your technique. This model will not encompass every athlete who jumps high, but will shed light on the model I feel is optimal for athletes learning to jump off of two feet for the first time. The key parts of a good two foot jump include but are not limited to: **the approach, the penultimate step, the arm swing, the plant foot, the**

block foot, and thee takeoff. Again, this is not an exhaustive list, but will serve to cover the key elements we feel lead to a good two foot takeoff.

Mastering the approach

The purpose of the approach or **run up** is to develop as much horizontal speed as possible and lower the center of gravity leading into the jump. In two foot jumping, there are many ways that an athlete can do this; some prefer to be rather slow and bouncy, while others like to sprint into the takeoff. Both of these styles can work, however athletes who are more elastic will be able to handle more approach speed. Keep in mind, these steps **do not** include the penultimate step. The most common error we observe during a two foot approach is the athlete not lowering themselves enough during these crucial lead up steps. As a result, they will almost always end up being very vertical leading into their penultimate step and unable to reposition themselves to maximize their takeoff. The athlete should feel like they are in an athletic position, moving smoothly from one step to the next. Again, **lowering** means bending the hips and knees. This should feel similar to an offensive player driving to the basket to finish at the rim. Lastly, it is crucial that the athlete maintain or even accelerate during this portion of the approach. The **worst** thing an athlete can do is slow down mid-approach. This will almost always detail the jump before it even starts.

It is also important to make sure that your **rhythm** starts relatively slow and builds progressively. The number of steps that the athlete should take varies depending on the circumstances. Volleyball players will have shorter approaches, while professional dunkers will

have longer approaches. For perspective, on Isaiah Rivera's 50.5" approach jump, he had five total foot contacts, including the penultimate step, plant, and block foot. This means the first two steps were solely focused on generating an initial velocity and lowering him prior to the contact of his penultimate step. In regards to the distance the athlete should start from the takeoff, the more speed and steps, the further away the athlete should be.

The last important variable to consider when approaching a jump is how much you plan to curve. This takes advantage of the lateral and rotational aspects of the jump as well as lowers the athlete more than they otherwise would have without the curve. The majority of the time this slight curve will help athletes **immediately** and it is recommended that athletes incorporate a slight curve into their approach.

Perfecting the penultimate step



If your goal is to have incredible jump technique, bookmark this section and re-read it several times. **The penultimate step will largely determine the success of a two foot jump.** I

have seen some of the most explosive athletes on the planet jump half as high as their weaker counterparts due to a poor penultimate step. Personally, it wasn't until I mastered this portion of the approach that I was able to actualize much of the progress that I'd made in the weight room.

Penultimate stands for second to last, and is arguably the most important part of the approach. It is where the athlete will lower their center of mass the most, and achieve their highest horizontal velocities into the jump. As previously mentioned, **how low you are and how fast you are** largely determine the potential you have to jump high. This is because the lower you are and the faster you are, the more you can load the stretch-shortening cycle in the legs, which ultimately explains how athletes are able to jump far higher than they could without an approach.

To execute a good penultimate step, you will want to contact the ground with your foot slightly in front of your hips.



This will inevitably lead to some braking forces, however it will allow you to balance appropriately in a low position. If you contact too far underneath the hips, you may begin falling

forward or find it difficult to reposition your limbs in an optimal position to jump high. If you contact too far in front of the hips, you will have excess braking forces and end up decelerating into the jump.

Once the foot makes contact with the ground, the knee should begin to roll forward and down.



This will allow the athlete to maintain a low position into their plant foot. If you rush this portion of the penultimate step, you will likely "**short**" the stride and end up with a somewhat jerky movement pattern.



Some athletes benefit from this, as they are better able to time the jump, but I much prefer a **long fluid stride**. Another cue that some athletes tell themselves to maximize the step is to actively push backwards on this stride. Personally, I have found that by doing this I generate a lot of speed and am very low, but have a tendency to shorten my push on the ground.

Another variable to consider when performing the penultimate step is the movement of the plant foot. Because the penultimate step happens in conjunction with the plant foot moving in front of the body, it is important to briefly touch on how the leg should move in front of the body. Ideally the plant foot is rotated somewhat outwards and swings through similar to a running stride. As the penultimate foot leaves the ground, the athlete should attempt to kick this leg out relatively straight.



This will allow them to convert their horizontal momentum vertically by generating high horizontal braking forces. These horizontal braking forces are also responsible for loading the

quadriceps during the jump, which, if the athlete is sufficiently strong, will result in a higher jump.

One commonly discussed element of the penultimate step is the **distance** of this stride. Because some athletes jump high with a short penultimate stride while others jump high with a long penultimate stride, there has been a lot of confusion around the topic. **That being said, as long as the athlete is low and fast, the length of the penultimate stride should not be overemphasized.** Further, the distance of the stride is determined by how far the penultimate foot moves behind the body, how low the athlete is, and how far the plant leg kicks out. If an athlete is very high and doesn't kick their plant leg out very far, their movement strategy will likely resemble a jump stop. **In contrast, if they are very low, leave their penultimate step far behind them, and aggressively kick out their plant foot, their movement strategy will resemble Ja Morant or Isaiah (Rivera).**

PS. The penultimate is the most important part of the approach but also the hardest to master! At THP we can analyze your technique and give you cues, tips, and drills to learn this skill. If you want us to help you with your technique sign up for our FREE 6 week challenge HERE

Maximizing the arm swing



One of the most underrated aspects of a good two foot approach jump is the contribution of an aggressive, well timed arm swing. The arm swing benefits two foot jumping by increasing the position of the center of mass at takeoff and by increasing the ground reaction forces. Because every action has an equal and opposite reaction, thrusting the arms upward results in a downward reaction force. This has the potential to add valuable inches to your vertical jump, **but only if timed correctly.** With that in mind, it is important that the arms begin moving forward and upwards as the block foot begins moving forward. As the hands move just in front of the hips, the block foot should contact the ground.



In the scenario where the athlete has a ball in their hand, he or she will want to thrust the ball upwards right as the block foot touches the ground.



If you do not do this correctly, you will not be able to convert horizontal momentum vertically correctly, and will have a very slow, disjointed takeoff. Another cue that I like to use to help athletes understand the timing is to imagine swinging your arms aggressively in a circle.

When you feel blood rush to the tips of your fingers most intensely is when the block foot should strike the ground.



When timed well, forces are optimally summated to maximize peak forces for a correctly sequenced takeoff.

One aspect not frequently discussed is the varying types of arm swings involved in an approach two foot jump. There are two commonly adopted styles of preparing for an effective arm swing; one is known as a **circular arm swing** while the other is known as the **pendulum arm swing**. In a **pendulum arm swing**, the hands move in front of the penultimate step in a clapping configuration.



This happens right before the penultimate step touches down and is followed by a synchronized backwards movement of shoulders, arms, and hands. This also perfectly syncs up with the movement of the penultimate step moving backwards. An illustration that we like to use to help athlete's understand this concept is imagining the hands and penultimate step being attached by a string. As the block foot (previously the penultimate step) begins moving forward, so do the hands. If you study the block foot and arm movement of elite two foot jumpers, you will always observe this movement sequencing. One potential downside to adopting this style is that while maximizing the aggression and contribution to vertical lift in the jump, it can also interrupt the maintenance of horizontal momentum.

As previously mentioned, the second commonly adopted arm swing technique is known as the **circular arm swing.** This arm swing is typically used by athletes who run very fast and tend not to think about their arm swing at all. It is merely the result of practicing hundreds and

hundreds of jumps at a high velocity. To do this correctly, imagine the movement of the arms in a running action. If you can picture this, you know that the opposite hand and opposite knee move in unison. The circular arm swing functions in a similar fashion, and explains why many athletes are able to carry a massive amount of horizontal momentum into the jump by using this type of arm swing. As the plant foot moves forward, the opposite arm moves forward. Similarly, as the penultimate step moves backwards, the opposite arm moves backwards. How much the athlete splits their legs during their penultimate step stride should be mirrored by the arm swing to ensure there is correct sequencing. As the athlete touches down their plant foot, the hand in front of the body moves upwards and backwards until it is in line with the other arm. At this point, the block foot and arms should be “on a string” and perfectly timed to maximize the jump. Again, the block foot should strike the ground right as the arms begin to feel the heaviest.



Handling the plant foot

The takeoff sequence in a two foot jump can be somewhat confusing due to the nomenclature adopted by many coaches and athletes, but the second to last **foot contact** is known as the plant foot.



Because penultimate means “second to last” it can be easy to misunderstand which is the plant foot and which is the penultimate step. That said, I like to view the plant foot and block foot as part of the **takeoff sequence.** With this nomenclature (or naming), it makes sense to continue calling the penultimate step as such.

The plant foot is the beginning of the takeoff sequence and is oftentimes an internal focus for new athletes trying to learn correct two foot technique. As previously mentioned, the plant foot is the aggressive outward kick that functions to brake the horizontal momentum generated in

the approach. Ideally, athletes should attempt to achieve a **long split** or a large distance between the toe off of their penultimate step and the touchdown of their plant foot. This oftentimes is associated with an aggressive push off of the penultimate step and high maintenance of horizontal momentum. It is also associated with a low center of mass position at a critical portion of the jump. If the athlete does not kick this leg out aggressively, it will feel like a small step. If they do kick this leg out correctly it will feel like a long fluid stride. If you have always taken a short penultimate stride, it may feel uncomfortable to lengthen this step, but I would encourage athletes to at least try to adjust their technique. If it does not lead to an improvement within a month or two, I would suggest going back to their previous technique.

It is also important to pay attention to the position of the athlete's torso during the flight phase of the penultimate step. At this point the athlete should have a very high horizontal momentum and as such will likely benefit from a vertical torso position.



This is not always the case, but very high level two foot jumpers do so to stop themselves from

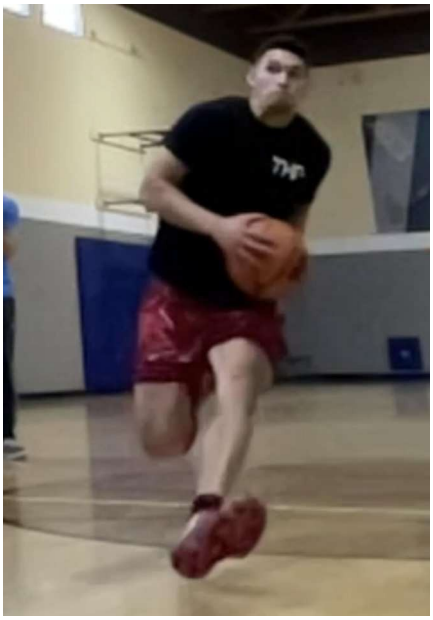
rotating forwards in the air. As an illustration, imagine running as fast as you can and jumping onto a trampoline with your shoulders too far forward; this would obviously result in you flipping forwards. Similarly, great two foot jumpers who have a torso that is angulated too far forward will inevitably rotate, or flip, that direction. This changes their highest touch point, making some dunks more difficult.

One interesting observation we've made from watching athletes perform approach jumps on force plates, is that the more acute the angle of the plant foot to the ground, the greater the horizontal braking forces. This means athletes are able to handle more horizontal momentum when they lift their knee and kick their foot upwards and outwards. This will likely feel like the athlete is **throwing** themselves into their plant foot.



One subtly that many athletes notice when watching elite two foot jumpers is the slight external rotation and inwards movement of their plant leg. This functions to further involve the

lateral hip musculature leading to a **whip and flail** mechanism. Because the plant leg should be relatively straight when contacting the ground, the pelvis is rotated backwards and upwards. This in turn rotates the opposite leg (the block foot) aggressively into the floor. Another benefit is that many athletes are much stronger in this configuration compared to having both of their feet facing forwards.



After the plant foot contacts the ground, the athlete should begin bending their knee in preparation for the contact of the block foot. At this point, the center of mass should continue to move forward and downwards. Stronger athlete's may have a tendency to flex their knees slightly more, but this is not always the case. The reason for this is that the muscles and tendons have differing properties that vary from person to person, such as fiber type, pennation angle, cross sectional area, and tendon stiffness.

Punching the block foot

The final foot contact prior to takeoff that functions to convert the remaining horizontal momentum vertically is known as the **block foot**. This foot should act like a whip, moving in unison with the arms, forward and downwards into the ground. The faster you run, the farther the foot should contact in front of the body. This is because, again, a more acute shin angle will lead to greater braking forces. Further, this foot should turn towards the midline.



This allows the athlete to further take advantage of the lateral hip muscles to effectively convert the horizontal momentum vertically. As an illustration, think about how much easier it is to walk down a very steep hill with the feet facing somewhat sideways as opposed to straight ahead. As the foot contacts the ground, the hands should be slightly in front of the body, and the athlete should be either volitionally or reactively working to push themselves upwards. The hands should drive upwards as aggressively as possible.

It is also important that this foot stay relatively low to the ground and whip forward in front of the body.



If done correctly the athlete's block side pelvis will rotate forward. This is inline with the previously discussed **whip and flail mechanism.** If this does not happen, the athlete is either not planting far enough across their body, or is running too slow to result in a true whip and flail mechanism. It is also possible that the athlete is bending their plant leg prematurely to offset the jarring sensation at touch down. If done correctly the block foot should slide into the ground as opposed to being set on the ground. As an illustration imagine aggressively sliding your foot into a flip flop. This will assist in getting the foot to move in front of the body at the correct angle.

Completing the jump: the takeoff

At this point in the approach two foot jump, most everything has already been determined. If the athlete is low, fast, and in the right position, he or she will have a good chance of jumping high. Early on, many athletes will benefit from focusing on actively extending the hip, knee, and ankle. This ensures that they are pushing through the entire portion of their jump, and as a result, maximizes impulse (and thus jump height). The other focus should be on driving the arms as high as possible to ensure the athlete is achieving the highest position of their center of mass at takeoff. It is also possible that this allows the athlete to generate additional reaction forces through the upwa



After you leave the ground, the trajectory of the center of mass is set, and there is nothing you can do to change its position in flight. Kicking the legs backwards, cycling the arms, or any other movement in the air will only result in a reactionary rotation elsewhere. As an example, long jumpers do this to stop themselves from rotating too far forward, but this is not a common problem that great two foot jumpers must overcome.

PS. At THP you can have your jump technique analyzed 24/7. You will get put in a group chat with our team of coaches where you can send them videos of your jumps. From there they will give you personalized tips and cues to instantly improve your technique (and as a result your vertical will go up instantly). Sign up for our FREE 6 week challenge HERE

ONE FOOT JUMP TECHNIQUE GUIDE



The purpose of this guide is to discuss the specific jump strategies for **one foot jumpers** for the purpose of creating vertical lift. In most cases, this applies to dunking and the track and field event, high jump.

General concepts

When jumping off of one foot, your ability to propel yourself off the ground is largely related to your **dynamic leg strength.** In research we classify this as the specific strength required by the takeoff leg to generate force over very short ranges of motion during a stretch shortening cycle. One caveat to this, is that one foot jumping is a very coordinated activity, and therefore should not be viewed in isolation, similar to repeat hopping on one foot. It is important to utilize the free arm and free leg at the correct time interval to effectively generate vertical lift.

How well you can take advantage of your takeoff legs dynamic leg strength is determined by two major variables: **how low your center of mass is** and **how fast your center of mass is moving at touchdown of the takeoff leg.**



If you are not low into the takeoff and you do not have a lot of speed, you will not be able to utilize the full capacity of your takeoff leg. One variable that will further complicate this is your nervous system's ability to turn muscles off and on at the correct time, at the right amounts, and in the right order. In research, we call this **synchronization,** and it largely dictates how well the muscle and tendon will be able to generate force in the takeoff.

One foot jumping

To master one foot jumping, it is important to understand the different focal points of a sound takeoff. Specifically, the athlete must master **the approach, the arm swing, the leg swing, the penultimate step, and the takeoff action.**

Mastering the approach

The purpose of the approach or **run up** is to develop as much horizontal speed as possible and lower the center of gravity leading into the jump. In one foot jumping, you will see athletes do this a few different ways. For high jump, athletes will run a curved approach. Similar to the way that a sports bike leans inwards to generate centripetal force so as to keep their tires connected to the road, athletes lean inwards to lower their center of mass without disrupting their running position prior to takeoff.



The second purpose of the curved run up is to allow for the generation of lateral somersault at takeoff, which we are not concerned with when looking at the one foot takeoff during dunking. Although dunkers do not want to generate somersault during their takeoff, they will want to focus on being as low as possible into the jump, in which case it can be useful to adopt a slightly curved approach into the jump. The second way in which athletes lower their center of mass is by bending their hip, knee, and ankle slightly during the final steps of the approach.



In high jump this typically happens during the curve and ensures that the athlete does not generate too much negative vertical velocity prior to touchdown of the takeoff leg. The more negative vertical velocity an athlete has, the more energy the athlete will have to spend on stopping the downward travel of their center of mass (a negative outcome when the goal is to produce positive vertical velocity). Because of this, it is important that the athlete lower themselves during the lead in steps and maintain this low position until the takeoff leg strikes the ground. The lower the athlete, the greater potential to load the takeoff leg at takeoff. The penultimate step, or second to last step will be discussed in the following section, but functions as a continuation of this process. Technically speaking, the penultimate step is also part of the approach, but we will discuss it in more detail as it is of greater importance than the approach.

While it is important to be low, the athlete will not be able to take advantage of the stretch shortening cycle without having a great deal of momentum into the jump. This brings me to the second objective of a good approach; the generation of horizontal velocity. The more horizontal velocity the athlete has, the greater the potential to load the takeoff leg at touchdown. In high jump, most athletes take a longer approach to generate a great deal of momentum during the straight portion of their approach, and then focus on the rhythm and setting up a proper takeoff during the final 3-5 steps. In dunking, most athletes will be limited to roughly 3 to 5 steps **total**. This means that to dunk, the athlete must generate most of their momentum in far less time. Further, dunkers have a markedly slower approach velocity than high jumpers. This distinction also explains why oftentimes athletes who can high jump well, struggle to dunk, as they don't have enough time, space, or energy to load their very stiff takeoff leg. Dunkers,

however, only need a few steps to set up a proper takeoff. The velocity of the approach in dunking should build relatively quickly, and the focus should be placed on spacing, taking the correct number of steps, and the rhythm. If spacing and the number of steps is correct, the focus should be placed on the rhythm building all the way through the takeoff. If the athlete focuses on speed too much, they will default to sprint mechanics and not maintain a low position into the jump.

In regards to rhythm, the cadence of the jump should accelerate all the way from the first step through the takeoff. In dunking, the greater the acceleration to and through the takeoff, the higher likelihood the athlete will jump high. The reason for this is that the stride frequency and not stride length will be recognized as the "rhythm" of the approach, and by having a very high stride frequency, the athlete will minimize flight times between each step, meaning they are maintaining a very low position. In many ways, the rhythm of the approach can be the greatest determinant of whether a jump will be successful or not. This is somewhat intuitive and must be learned through spending countless hours practicing good jumping technique.

Perfecting the penultimate step

Arguably the most important step of the entire approach is the second to last stride, or penultimate step. The reason this stride is so important is that the energy generated in the previous steps can either be preserved and improved upon to maximize the jump height, or entirely dissipated, making every step prior to this one essentially useless. This stride can be performed a number of ways but major things must be preserved in order to perform a successful

penultimate step: 1. The athlete must maintain a low position into the takeoff. 2. The athlete must maintain as much velocity as possible from the previous stride. To the first point, it is better to maintain a low position or push slightly upwards into the plant leg than to continue to lower during this stride. If the athlete continues to lower, they will likely have a very high negative vertical velocity prior to touchdown of the plant foot. Secondly the athlete must minimize braking forces during this stride.

To achieve these big picture goals, the athlete should aim to perform a rolling foot action with the shin relatively vertical at touchdown.



The more the athlete plants in front of the body, the more deceleration will take place. This interferes with the second goal of a good penultimate step, maintaining horizontal velocity. As the athlete moves through this step, their knee should move downwards and roll towards the

ground.



From mid support to toe off, if possible it is better to push the penultimate step through the ground, moving the hips forward and shoulders back. Contrary to popular belief it is during this stride that the athlete determines whether he or she will be in proper position at touchdown of the takeoff leg. By rolling through the penultimate step correctly, the athlete will end up with their glute relatively contracted during the final push off, thus pushing the pelvis forward and allowing for optimal braking and thus transfer of horizontal to vertical momentum.



As far as lowering, the athlete should stride to keep the shoulders tall or even back and perform a rolling step while in a semi squatted position. If done right it will likely feel like a low walking lunge. As the athlete performs this step the takeoff leg should move in front of the body, sweeping through low to the ground in preparation of takeoff. The arms should move from a clapping position in front of the penultimate step to thrust behind the body. Some speed jumpers will perform a running arm action. It is important to note that the athlete should not jump into the penultimate step, nor disrupt the progressively faster rhythm of the run up during this stride. As easy as it may sound, maintaining how low you are, how fast you are, and ensuring that your rhythm is progressively increasing is very difficult, as the position of the athlete will change a great deal during this step.

Maximizing the arm swing

The arm swing that an athlete chooses for a one foot jump will likely coincide with the arm action that he or she selects. Because the athlete must generate peak forces in unison or in a synchronized manner, he or she will want to mirror the leg action with the arm action. If an athlete uses a very short running arm action with a very long and slow leg drive, he or she will come off as discoordinated. Similarly, if an athlete uses a very large arm swing with a very short leg swing, he or she will also jump lower than the scenario where the arm drive and leg swing were of similar amplitude and velocity.

The two major arm swings that an athlete will default to when it comes to one foot jumping is a running arm action or a pendulum arm swing. In a running arm action, the arms

move in opposition to the legs, similar to running (hence the name). For example, if the right knee is the drive knee, then this knee will be moving upwards while the left arm is moving upwards. **During the penultimate step in a left leg jump,** the athlete should strive to have their opposite arm in front of their penultimate step. As they roll through that stride, and the takeoff leg moves forward, as does the opposite arm. This will almost feel like a twisting motion if done correctly. As the left leg hits the ground, the right hand should be in front (and across) the body. The left arm should be behind your left hip. As the left leg hits the ground, the right will drive laterally and downwards as the right knee punches upwards. The left hand will punch upwards and across the body in unison. This will create a very fast takeoff. If your goal is to attempt to speed jump, I would encourage you to try this arm action, as it will shorten the ground contact time.

The second way to perform the penultimate step is to utilize a pendulum arm swing. During this arm action, the hands are moved in front and over the penultimate step in a clapping action. As the athlete rolls through their penultimate step, the hands move backwards and downwards. The takeoff leg moves forward as the penultimate step moves behind the body. At the same time, the athlete should have his or her arms as far behind them as possible. This arm swing is much slower, but is considered to be the strongest arm swing as it pertains to generating vertical lift. For this reason it is the most effective, however, if the athlete is better at generating force over shorter periods of time, he or she may want to consider a running arm action. Most power jumpers utilize this takeoff. One downside to this style of arm swing is that it disrupts the maintenance of horizontal momentum during the penultimate step and can lead to braking and thus less potential to jump high during the takeoff action. As the athlete plants the takeoff leg, he or she should swing the arms in unison with the swing leg. If done correctly, as the athlete

reaches the deepest knee flexion of their jump, the knees should be stacked and the hands should be just in front of the drive knee.

The drive knee (Speed jumping vs. power jumping)

Contrary to common knowledge, power jumping and speed jumping has very little to do with the run up speed. Power jumpers can have a higher takeoff velocity than speed jumpers and vice versa. What determines whether an athlete is a power jumper or speed jumper is what happens during the knee drive. As an athlete toes off of the penultimate step, the foot can either stay low to the ground and swing through with a very large moment of inertia, or the foot can move upwards and begin to recover in a running action towards the butt with a very small moment of inertia. When it comes to the strength of the leg drive, the large moment of inertia is considered to be stronger and thus better for generating vertical lift. Despite this, many athletes stride to have a short moment of inertia as it is better for high jump, coincides with shorter ground contact times, and allows for a faster conversion of horizontal to vertical momentum. It is important to note that **both strategies are effective for producing vertical lift.** Oftentimes athletes with a soccer background will adopt a speed jumping free leg, whereas basketball players will adopt a power jumping free leg. The reason for this, in my opinion, lies in the fact that soccer players are able to swing their free leg through in a similar manner to speed jumping whilst striking a ball, whereas basketball players are forced to respond to defenders and control a ball during a jump. The latter requires the athlete to slow the takeoff down, resulting in a slower takeoff strategy. If you're reading this and have been jumping for some period of time, which jump strategy you select is likely already predetermined. I would not suggest trying to force yourself into either category, however if you would like to attempt to learn speed jumping, I

would encourage you to practice it with your other leg, so as to not disrupt the natural coordinative tendencies of your dominant takeoff leg.

In regards to the actual leg swing, the athlete should strive to drive the knee in unison with the arms, mimicking the speed and timing of the arm drive. The height of the knee drive should not be overemphasized as this can disrupt the coordination of the takeoff. It is more important that the leg drive match the arm drive as well to optimize force summation during amortization of the takeoff leg. If done correctly, the takeoff will feel like a swift kick in the ass that throws you into the air.

The takeoff leg

Now that we have covered the approach, the penultimate step, and the arm and leg drive, we can discuss the ever important takeoff. If each of the previous steps has been completed correctly, it is almost impossible to have an unsuccessful takeoff. This is because the athletes velocity and center of mass should already be predetermined from the previous steps. If the athlete toes off the penultimate step with the torso upright, the hips forward, and the arms back, all the athlete must do is plant their takeoff leg in front of their body. The further the foot is placed in front, the more braking forces will take place. It is encouraged that the athlete plant their foot across their body, slightly towards the midline, whilst leaning slightly inwards and backwards. The athlete will want to plant their foot on the side of their foot or on the outside of their heel. If you observe high jumpers, some will plant on the side while others will plant on the

back and side of their heel. This is somewhat of a preference.

Depending on what type of jumper you are, power (toe dragger) vs speed (butt kicker), you will want to focus on a slightly different type of sensation. Speed jumpers typically like to focus on having the fastest and punchiest takeoff possible, while power jumpers will want to focus on having a smooth long draw on the ground. If you are a high jumper, you likely will want to default to the former, even if you are a power jumper. The more velocity you have, the shorter and faster the takeoff should feel. The slower you are, the longer the takeoff should feel. I like to use the analogy of throwing different types of punches to help illustrate this point. A faster jumper will feel like they are throwing a jab into a heavy bag, whereas a slower jumper will feel as though they are winding up to throw a haymaker. Through trial and error and using your jump height as a guide, you will innate begin to feel which cue works best for your individual jump style.

The athlete **should not focus on how much to bend their knee** as this will slow the takeoff down, regardless of what style of jumper you are as well as disrupt the natural coordination of the takeoff. It will impact precontraction, rate coding, synchronization, and the fluidity of the movement pattern. The athlete's knee will bend according to their dynamic leg strength, tendon properties, and their technical proficiency.

Torso and hip position

One very important aspect of a successful one foot takeoff is the position of the takeoff leg and torso during touchdown of the takeoff leg. Many coaches will not consider a takeoff

successful if the athlete does not have correct alignment of the takeoff leg through the same side shoulder during touchdown of the takeoff leg. If done correctly, when viewed from the side, you should be able to draw a straight line from the athlete's heel all the way through their same side shoulder. This is one of the most difficult things to achieve, but likely comes through practicing max effort takeoffs where peak jump height is the emphasis. As someone who has spent 17 years perfecting their one foot takeoff, I have found that focusing on this typically leads to errors and disruptions in rhythm, acceleration, and aggression of the takeoff, however, is often achieved through having a fast rolling penultimate step where my penultimate step achieves peak extension at the hip. Because the pelvis is connected, this will move the takeoff leg hip forward, and if the shoulders stay tall or even slightly backwards, will result in perfect alignment at touchdown of the takeoff leg.

The torso position of a one foot jumper will also greatly impact the success of the jump. The athlete should strive to stay tall during the takeoff. It is okay, and even advised, that the shoulders stay forward in acceleration positions during the lead in steps (of a dunk particularly), but as the athlete contacts and rolls through the penultimate step, **it is imperative that the shoulders move behind the hips.** If done correctly it will feel as though the athlete is leaning backwards at the shoulders as their hips move forward into the plant. I would encourage you to practice walking through this motion if you cannot conceptualize what this feels like.

We hope this technique guide helped you jump higher! If you want MORE help with your jump technique sign up for our FREE 6 week challenge HERE

The Science Behind Tendon Pain (And How To Get Rid Of It)

PS. If you want to skip the science and go straight into the sets and reps you can skip to the end (not recommended, as understanding the science helps you do the workouts correctly).

If you want to watch the video version of this guide (where give away even more secrets and go more in depth on the programming) click here: https://youtu.be/Vpy3zuT5yQo

What Are Tendons

- Tendons connect muscle to bone
- Think of them like biological “springs”
- Their job is to transfer energy from muscles into movement (jumping, sprinting)
- They do this quickly by storing and releasing elastic energy
- HEALTHY TENDONS DO NOT TEAR
    - They are incredibly strong and built to handle massive amounts of load

How Tendons Function

- They stretch to store energy
- They recoil to release energy (just like a spring)
- The reason this is so valuable for running and sprinting is because they can do it extremely quickly. For example, in sprinting, the Achilles stores and releases energy in as little as 80 milliseconds of ground contact time.
- **Viscoelasticity**: Tendons can be both compliant (stretchy) or stiff depending on how much load is going through it and how quickly it’s being applied
    - This is governed by Hooke’s Law: Force = kx where k is the stiffness of the spring and x is how far the tendon stretches
    - The stiffer the tendon and the farther you stretch it, the more force you apply when this energy is released.

What Determines Tendon Load?

1. Reps: How many times the tendon stores and releases energy
2. Speed: How quickly the tendon is is storing and releasing this energy

3. Joint Angle: The more the tendon is stretching under the load the higher the stress on that tendon

Higher load = HIGHER RISK (this is why more explosive athletes experience more tendon issues)

The Tendon Pain Continuum

- Based on Jill Cook's research

1. Normal Tendon: Strong and healthy. Good at bearing load.
2. Reactive Tendinopathy: Short term reaction to getting overloaded. Pain usually happens when you spike intensity or volume over a short period of time.
3. Tendon Disrepair: This is when structural changes start to occur in the tendon, but not enough that it's not reversible.
4. Degenerative Tendon: Permanent cell death. Not as bad as it sounds because you can still build strong, healthy tendons around the degenerated cells.

KEY POINT: Structure does not equal pain! I REPEAT. STRUCTURE DOES NOT EQUAL PAIN. Some athletes have extremely degenerated tendons with no pain, and others feel pain with barely any degeneration in the tendon.

Why Jumpers & Sprinters Are Vulnerable

- Your tendon undergoes massive amounts of load when you jump and sprint.
- During sprinting you can experience up to 6x your body weight on a single leg.
- During one-foot jumping it can be up to 14x body weight.
- Dunkers are especially at risk because you're undergoing deep joint angles, insane amounts of force, happening very quickly, and you're doing it over and over again. Jump sessions can last upwards of two to three hours. Which is a recipe for inducing tendon pain.

What Does NOT Work

- Stretching the tendon adds more load to an already stressed tendon.
- Ibuprofen or anti-inflammatories. Most tendon issues are not inflammatory.
- Random eccentric exercises. This often just overloads a tendon beyond its capacity and induces pain.
- Icing the tendon. Again.. Tendinopathy is not an inflammatory issue.

These are just band aids and NOT the solution!

What Actually Works

- Load management is the only proven long-term approach.

Good load management is:

1. Reduce pain first
2. Progressively reintroduce load to increase the tendons capacity.
3. Then reintroduce sprinting and jumping safely and slowly.

This is NOT about fixing structure. It's about improving function (what you can do without feeling pain).

Here is what our load management protocols look like (the secret sauce).

Phase 1: Isometric Loading

- An isometric is any time a muscle is undergoing load without changing its length.
- Prescription: 5 sets of 45 seconds at 70% of your maximum voluntary contraction. Repeat three times per day. Repeat every 6 hours.
- Science
    - Isometrics have an analgesic effect, meaning that pain decreases. I like to think of them like ibuprofen. Doing isometrics can reduce pain for up to 90 minutes after the 5 sets.
    - Isometrics also restore normal muscle activation patterns that were inhibited by tendon pain before.
    - It's also a way for us to load the tendon in a controlled way (while avoiding the two things that are most provocative: speed and length)

Phase 2: Slow Heavy Strength Loading

- What is it: Going slow tempo on the eccentric AND concentric part of a strength exercise
- Prescription: 3-4 sets of 6-8 reps. 4 seconds down, 4 seconds up. Every other day
- Science
    - Increases tendon stiffness.
    - Increases strength in the muscle.
    - Going slow reduces stress on the tendon while making it adapt to become stronger.

    - Progressively overloading by adding 5-20 pounds to the exercise per session allows the tendon to tolerate the load without feeling pain. AKA Capacity goes up.

Phase 3: Energy Storage

- What is it: Depth lendings progressing from low heights to higher heights over time.
- Prescription: 4 sets of 6-8 drops.
- Science
    - Trains the tendon to absorb forces that develop more rapidly than what you see in strength exercises.
    - Prepares athletes for being able to release that energy in a later phase.
    - Here the tenant starts to develop enough capacity to play sports.

Phase 4: Energy Storage AND Release

- What is it:
    - Standing jumps
    - Approach box jumps
    - Short approach jumps
    - Long approach jumps
- Prescription: Sessions should be about 20 jumps and increase in intensity over time. Once you can handle max intensity, you can then start increasing the length of the sessions over time.

Too Much Information To Process? Let Us Take The Guesswork Out Of Your Training

Periodization is at the heart of every good program. At THP we build periodization into every cycle. You will literally never do the same week of training twice. This guarantees that you are always adapting and pushing to get better. Sign up for our FREE 6 week challenge HERE